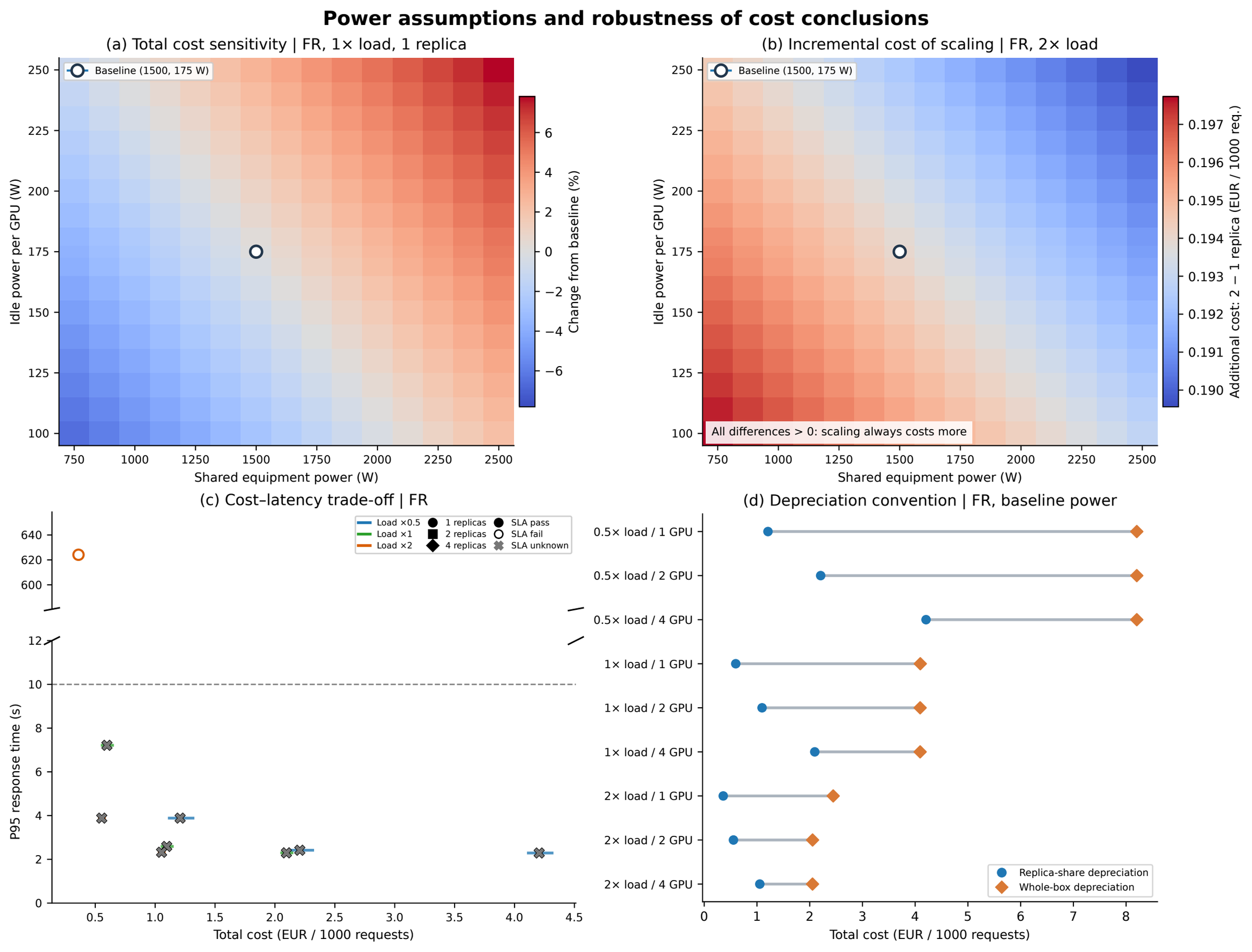

Power assumptions and robustness of cost conclusions
(a) Total cost sensitivity | FR, 1× load, 1 replica
Baseline (1500, 175 W)
Idle power per GPU (W)
Shared equipment power (W)
Change from baseline (%)
(b) Incremental cost of scaling | FR, 2× load
Baseline (1500, 175 W)
All differences > 0: scaling always costs more
Idle power per GPU (W)
Shared equipment power (W)
Additional cost: 2 − 1 replica (EUR / 1000 req.)
(c) Cost–latency trade-off | FR
Load ×0.5
Load ×1
Load ×2
1 replicas
2 replicas
4 replicas
SLA pass
SLA fail
SLA unknown
P95 response time (s)
Total cost (EUR / 1000 requests)
(d) Depreciation convention | FR, baseline power
0.5× load / 1 GPU
0.5× load / 2 GPU
0.5× load / 4 GPU
1× load / 1 GPU
1× load / 2 GPU
1× load / 4 GPU
2× load / 1 GPU
2× load / 2 GPU
2× load / 4 GPU
Replica-share depreciation
Whole-box depreciation
Total cost (EUR / 1000 requests)